

LMS Instructions

Submission Format -

Download your notebook as .html or .pdf and compress to zip folder to make a submission and make sure all outputs and visuals are seen in the submitted file.

Evaluation Metrics -

Step	Purpose & Key Expectations	Points
1. Understand Your Data	Load, preview, and document the dataset; check data types, missing values, duplicates; distinguish categorical vs. numerical fields.	10
2. Descriptive Statistics	Produce accurate summary metrics (mean, median, std dev, min-max for numeric; mode/frequency for categorical) and comment on central tendency & spread.	15
3. Data Visualization	Create well-labeled charts (histograms/boxplots, bar charts, scatter/KDE plots) that reveal patterns and include brief insights.	15
4. Bivariate Analysis	Run correlations, crosstabs, and grouped summaries to explore relationships between variables and highlight noteworthy patterns.	15
5. Formulate Hypotheses	State clear null & alternative hypotheses for each business question; select appropriate statistical tests and list their assumptions.	10
6. Run Hypothesis Tests	Execute the chosen tests, verify assumptions, report statistics & p-values/confidence intervals, and interpret results in business terms.	20
7. Present Business Insights	Translate findings into concise, data-backed recommendations with quantified impacts and clear visual or written presentation.	15